Information Seeking Across ChatGPT, Google Search and YouTube

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We investigate how people distribute information-seeking tasks across generative AI and web search. We analyze complete user-donated account histories from over 200 users across ChatGPT, Gemini, Google Search, and YouTube using a unified intent taxonomy and chronological tracking. We present four contributions. First, platforms specialize by functional intent, not topic. Second, individual users consistently maintain this division, directing complex reasoning and artifact generation to conversational assistants, procedural tasks to video platforms, and factual or navigational lookups to search engines. Third, users routinely bridge these isolated platforms through manual handoffs to complete single, multi-stage tasks. Finally, through an event-study panel analysis, we show that following ChatGPT adoption, a user's web search volume declines by approximately ten percent, but their underlying search composition remains completely unchanged. Generative AI absorbs new expressive capabilities rather than substituting traditional search.

CCS Concepts: • **Human-centered computing** → **Empirical studies in HCI**; • **Information systems** → **Web log analysis**; *Users and interactive retrieval.*

Additional Key Words and Phrases: generative AI, web search, information seeking, data donation, search intent, ChatGPT, YouTube, India

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1 Introduction

When people engage in online information seeking today, they interact with a few distinct surfaces. Rather than relying solely on traditional keyword-based web search engines like Google, users increasingly turn to conversational generative AI interfaces such as ChatGPT or Gemini, as well as video search platforms like YouTube. Each of these surfaces represents a fundamentally different interaction paradigm: Google Search for general web retrieval; conversational assistants provide interactive dialogue and synthesized artifacts; and YouTube (the second largest search engine globally [42]) serves as the primary destination for visual, procedural, and multimodal retrieval. Most people who use one of these tools regularly incorporate the others into their daily habits [26].

Despite this shift, human-computer interaction research still lacks an empirical account of how users actually divide their information-seeking tasks across these coexisting platforms in everyday life. We have very little behavioral evidence showing which specific types of requests go to which interface. We also do not fully understand how people coordinate these systems when working through complex goals. Public and academic debates constantly question whether generative AI will displace, cannibalize, or complement web search [1, 9, 15, 31, 44]. However, our empirical models lack a baseline description of how users navigate this ecosystem when all three options are readily available.

Understanding how people distribute their information seeking across platforms is important because interface affordances directly shape user sensemaking, cognitive authority, and the sustainability of the web. Information seeking is one of the most pervasive human-computer interactions in modern life [23, 30]. When a user routes an information need to a web search engine, the system returns a ranked list of independent web sources, placing the burden of information foraging, credibility assessment, and synthesis on the user [2, 30]. In contrast, when the same need is directed to a conversational assistant, the interface delivers a pre-synthesized, single-voice response that largely

obscures source provenance and eliminates exploratory browsing [1, 37]. Meanwhile, video platforms provide rich procedural and embodied demonstrations that neither text interface can match. Where a user directs an information need determines what kind of answer they receive, how they assess its trustworthiness, and which creators and publishers receive engagement. Furthermore, identifying the true distribution of requests clarifies what each interface must be designed to support: an assistant routinely consulted for personalized life advice requires radically different transparency, conversational memory, and safety guardrails than a search engine optimized for rapid navigational lookups.

From a conceptual standpoint, foundational HCI and information retrieval frameworks were established when information seeking was synonymous with finding and reading web documents. Classical models, such as Bates' berrypicking [2], Kuhlthau's information search process [24], and Marchionini's exploratory search [30], as well as standard query taxonomies [5, 33], were formulated around keyword queries typed into a search box. Today, the discrete steps of an information-seeking task are routinely distributed across systems that generate text, index documents, and stream video. Before the field can update its conceptual models of interactive retrieval, we need an empirical account of how users coordinate their information seeking across these boundaries.

Prior empirical studies have captured only isolated facets of this phenomenon. Server-side log studies analyze prompts or queries at massive scale, but are completely blind to off-platform behavior [7, 10, 22, 45]. Passive web browsing panels track cross-domain traffic over time, but log only domain visits and URLs rather than actual query text or prompt contents [3, 15, 18, 31]. Surveys and interview studies provide valuable insights into user perceptions and stated motivations, but are inherently limited by recall bias and post-hoc rationalization [26, 28, 41]. Finally, prior taxonomies were developed for single paradigms: web search taxonomies cannot classify generative prompts, while prompt taxonomies cannot categorize keyword lookups, preventing direct cross-system comparison [4, 25, 36, 40].

We address these methodological hurdles through a mixed-method data donation approach. Rather than relying on single-platform logs or coarse browsing panels, we analyze complete, user-donated account histories from the same individuals in India across ChatGPT, Gemini, Google Search, and YouTube, capturing the exact text and timestamps of every interaction. We apply a unified taxonomy developed by Lichtenegger et al. [27] to classify every request by topic (across ten domains) and functional intent (across eight categories). Applying a shared classification scheme allows us to measure cross-platform routing directly, both across cohorts and within individual users. To observe how information-seeking tasks unfold across platform seams, we complement our quantitative analysis with qualitative close readings of fifteen complete user trajectories in chronological order. Finally, we conduct an event-study panel analysis to examine how an individual's Google Search volume and composition change following their adoption of ChatGPT.

Our findings show that generative AI reorganizes information seeking rather than displacing web search move for move. Specifically: (i) Platforms differ by functional intent rather than topic. When an information need could be served by either system, users make consistent choices, routing factual lookups to web search, conceptual explanations and personal advice to conversational assistants, and sharing instructional tasks across assistants and video platforms. (ii) This functional division holds consistently within individual users. Controlling for overall activity, an individual is 36 times more likely to direct an advice request to ChatGPT than to Google Search compared to a factual lookup. (iii) Participants routinely chain these systems together within single tasks. Because current systems offer no native support for cross-platform workflows, users manually carry context across tools, such as drafting a plan in ChatGPT and then searching Google for the specific tools it recommended. (iv) Following a user's first recorded conversation with ChatGPT, the fundamental composition of their Google searches remains virtually unchanged. While total search

volume experiences a modest ten percent decline, generative AI ultimately absorbs new expressive tasks while leaving the core functions of search entirely intact.

These findings demonstrate that assessing generative AI's impact on search purely through aggregate query counts obscures the structural reorganization taking place. By showing how users divide intent across modalities and navigate the friction between them, this work offers empirical grounding for future HCI research on cross-surface information seeking and the design of next-generation retrieval interfaces.

2 Related Work

Our work builds on four areas of prior research: models of information seeking as an interactive process, empirical studies of conversational AI, research on the interaction between generative AI and the web, and the theoretical frameworks and intent taxonomies used to understand media coexistence.

2.1 Information Seeking as an Interactive Process

Research in Human-Computer Interaction and information science has long shown that information seeking is an iterative process rather than a set of isolated query-response events [43]. In her classic berrypicking model, Bates [2] showed how searchers continuously reformulate their needs as they encounter new information, while Kuhlthau [24] demonstrated that uncertainty and cognitive shifts guide user strategies over time. Marchionini [30] separated simple factual lookups from exploratory search, where learning and synthesis unfold across prolonged interactions, and Kellar et al. [23] found that distinct online goals leave clear traces in session length and navigation.

A central insight of this literature is that the true unit of analysis is the user's overarching task rather than the individual query [6, 20]. Much of this activity occurs during everyday life information seeking, spanning health, financial, and personal decisions [34]. Furthermore, information seeking extends well beyond text search: video platforms like YouTube have become major search engines in their own right, especially for procedural, instructional, and visual tasks where text alone is inadequate [42]. Because an information-seeking task consists of multiple stages, users can naturally distribute distinct sub-tasks across tools, delegating some moves to an assistant, some to web search, and others to video. Consequently, measuring technology adoption purely by counting queries obscures how the broader task is arranged.

2.2 Generative AI in the Wild

A second body of work examines how people interact with conversational language models through interaction logs. Early public releases such as WildChat [45] and LMSYS-Chat-1M [46] enabled initial observations of user prompting in the wild. Large-scale platform studies soon followed: Chatterji et al. [7] analyzed hundreds of millions of weekly ChatGPT conversations, finding that workplace tasks accounted for only 27% of interactions, while personal curiosity and creative expression grew rapidly. Complementing platform-side data, donation studies observe complete account histories directly from consenting users. Karnam et al. [22] documented growing anthropomorphic framing and requests for mental health advice, while Chowdhury and Garimella [10] analyzed donated conversations across India, Nigeria, Brazil, and Pakistan, finding that personal problem-solving, health guidance, and translation outweighed workplace tasks. While these studies provide rich detail on how people use chatbots, they observe a single platform in isolation. They cannot reveal what those same individuals searched for elsewhere, nor how they split work between tools.

2.3 Generative AI and Web Search: Substitution vs. Complementarity

A growing body of research examines how generative AI affects web browsing and search, though empirical findings remain mixed. Using desktop browsing panels, Gholami et al. [15] observed that ChatGPT adoption coincided with an increase in unique websites visited and a temporary rise in search, whereas Padilla et al. [31] documented a steady decline in conventional search volume alongside traffic losses for reference and educational sites. In domestic settings, Blank et al. [3] found that adoption increased leisure browsing while productive time remained unchanged, and field experiments with Google AI Overviews show that synthesized summaries reduce outbound link clicks by 38% [1].

Other work explores how users divide labor conceptually between these interfaces. Yu et al. [44] found that nearly 80% of ChatGPT requests could not be handled effectively by conventional search engines, and Suri et al. [38] showed that generative search better supports complex knowledge work. Qualitative studies indicate that people mix and match tools based on perceived task risk and complexity [26, 41], using AI for drafting and synthesis while keeping web search dominant for factual grounding [19, 28]. Closest to our design, Iannelli and Ai [18] linked prompt data to browsing logs in the US and UK to reconstruct multi-tool sessions, finding that journeys typically begin in search and transition to an assistant. However, their data captured domain-level visits and timestamps rather than the semantic text of queries and prompts. Prior work thus measures high-level traffic or relies on retrospective self-reports; none captures the full text and chronological sequence of requests that the same individuals direct across search, chat, and video platforms.

2.4 Cross-Platform Intent and Media Coexistence

Comparing information seeking across platforms requires both a theoretical understanding of media coexistence and a shared taxonomy to measure it. Theories of media competition suggest that new technologies rarely displace incumbents wholesale. Dimmick's Theory of the Niche [12, 13] posits that a new medium displaces an older one only where both satisfy identical user gratifications; otherwise, they undergo niche differentiation and coexist. Similarly, Dutta-Bergman's Channel Complementarity Theory [14] shows that strong interest in a topic leads individuals to use multiple channels concurrently, while work on media repertoires [17] and polymedia [29] emphasizes that users compose personalized suites of tools based on functional affordances.

To evaluate these dynamics empirically, researchers need a common description of user intent across interfaces. Traditional web search taxonomies, beginning with Broder's [5] distinction between informational, navigational, and transactional queries, and refined by subsequent work [21, 25, 33] were built specifically for short keyword queries [32]. Conversely, conversational taxonomies [4, 36, 40] focus on dialogue, synthesis, and creative generation. Because these schemes are paradigm-specific, they cannot be set side by side.

We resolve this tension by adopting the unified taxonomy proposed by Lichtenegger et al. [27], which spans search engines and conversational chatbots within a single framework across three families (Knowledge, Guidance, and Output) and eight functional categories. This unified classification allows us to test the concrete predictions of niche and complementarity theories: namely, that any displacement will appear within the specific functional intents that both search engines and assistants can serve, that topical interests will correlate positively across platforms, and that routing preferences will reflect consistent individual strategies rather than random variation.

3 Data and Methods

To study how information seeking is distributed across platforms, we collected an account-level data donation corpus and classified each request by both subject matter (topic) and functional goal (intent).

Source	Users	Records	Unit	Span
ChatGPT	300	41,054	conversations	Dec 2022 – Feb 2026
Gemini	232	8,061	conversations	Jul 2024 – Feb 2026
Google Search	289	922,901	search queries	Sep 2007 – Feb 2026
YouTube Search	248	404,666	search queries	Dec 2010 – Feb 2026
YouTube Watched	291	5,892,952	video titles	Dec 2010 – Feb 2026

Table 1. The corpus by source across all donors and years. Cross-platform comparisons are restricted to the paired cohort across the aligned window (January 2023 to February 2026).

3.1 Data Donation and Corpus

We collected a cross-platform data donation corpus from consenting participants in India recruited through Clickworker [11]. Participants exported and donated their complete ChatGPT conversation histories using OpenAI's export functionality, alongside their personal Google activity archives generated via Google Takeout. From the Takeout archives, we extract Google Search queries,[1] YouTube search queries, YouTube watch history, and Gemini conversational activity (Table 1). We parse each source to identify the core information-seeking request: for ChatGPT and Gemini, we treat each multi-turn conversation as a unit (represented by the initial user prompt); for Google Search and YouTube Search, each record is an individual search query; and for YouTube watch history, each entry represents a watched video title.

To enable meaningful cross-platform comparisons, our main analyses focus on a paired cohort of 209 participants who donated both ChatGPT and Google Search archives over an aligned window from January 2023 to February 2026.[2] Our participants are platform users residing in India, and have a median age of 24 (interquartile range 21–29) and 83% identify as men.

Ethics. This study was reviewed and approved by our institution's IRB. Participants went through an extensive consent process detailing the data collection and usage and were compensated for their time. Participants chose which archives to export and could withhold any source. To safeguard participant privacy given the sensitive nature of personal search and conversational logs, all classification and parsing pipelines were executed entirely on local, access-controlled institutional servers using open-weight models, ensuring that no participant data was transmitted to external third-party APIs.

Positionality. Our author team includes researchers who grew up in India and are fluent in Hindi and romanized Hindi, as well as researchers based internationally. Linguistic and cultural familiarity was essential for interpreting localized references such as government exam acronyms, state welfare schemes, devotional media, and code-mixed Hinglish queries. At the same time, we acknowledge our own professional distance from the economic circumstances of crowdworkers, which bounds our interpretations to what people asked and watched rather than their unstated motivations.

3.2 Topic Classification

Every request is categorized into one of ten broad topics: Technology & Computing, Education & Learning, Entertainment, Finance & Business, Health & Wellness, Content & Social Media, Personal & Lifestyle, News & Society, Writing & Language, and Miscellaneous. We inductively derived these categories from our data, building upon recent taxonomies

[1] Google's generative search features (such as AI Mode) were not broadly rolled out in India during the period of our study. The Google Search logs in our corpus therefore capture traditional web search queries rather than conversational generative search sessions.
[2] Depending on the specific analysis requirements (such as needing pre-adoption logs or all five platform archives) the exact number of participants varies; see Appendix Table 3 for a detailed breakdown.

of cross-platform information seeking and conversational AI usage [27, 44] to ensure they spanned the full spectrum of requests across all three surfaces. Because interaction formats differ fundamentally across tools, we employ two complementary classification pipelines.

For conversational sources (ChatGPT and Gemini) and YouTube watched video titles, we cluster requests using BERTopic [16] on Qwen3-8B embeddings. UMAP and HDBSCAN parameters were tuned per platform using Optuna to maximize cluster coherence and minimize unassigned outliers (Appendix Table 4). This produced 200 semantic clusters for ChatGPT, 236 for Gemini, and 68 for YouTube watch history. We then prompted Qwen3.6-35B to generate descriptive labels from representative cluster documents, inspected them manually, and grouped them into the ten top-level topics.

For Google Search and YouTube Search, queries are too short and syntax-compressed for dense semantic clustering. We instead classify search queries directly into Google Trends categories [8, 39] using Qwen3.6-35B in a two-stage prompt (predicting a top-level domain followed by a subcategory. See the full prompt in Appendix B.2). These categories were then mapped onto the same ten overarching topics. Because the Google Trends taxonomy lacks a dedicated category for text editing and does not generate unclassifiable records, search queries are structurally never assigned to Writing & Language or Miscellaneous.

3.3 Intent Classification and Validation

While topic captures what a request is about, *intent* describes the functional move the user expects the system to perform. We adopt the cross-platform taxonomy of Lichtenegger et al. [27], which organizes user goals across search engines and conversational assistants into three families and eight categories:

- **Knowledge:** *Concrete* (verifiable facts or definitions, e.g., "what is plant larva soil"); *Conceptual* (explanations of mechanisms, e.g., "difference between kernel and shell in operating system"); and *Contextual* (time- or location-sensitive information, e.g., "weather today").
- **Guidance:** *Instructional* (procedural how-to steps, e.g., "how to use hair minoxidil"); *Evaluative* (comparisons across alternatives, e.g., "living in Latvia versus US"); and *Strategic* (personalized advice requiring judgment, e.g., "should I go to this retirement party").
- **Output:** *Access* (navigating to or operating an external resource, e.g., "facebook", "mic test"); and *Deliverable* (generating a finished digital artifact, e.g., "update my resume", "write a birthday poem").

Three of these intents are constrained by platform affordances: chatbots cannot navigate to external URLs (Access), while standard search engines do not compose customized text artifacts (Deliverable) or monitor real-time physical conditions (Contextual). The remaining five intents can be served by either system, making them the primary locus of user choice. We classify every request using Qwen3.6-35B using the prompt provided in Appendix C.

To validate classification quality, we manually coded label-stratified samples of 200 requests from each of the five sources (2,000 items in total). The intent classifier achieves 84.7% overall accuracy relative to human ground truth (Cohen's $\kappa = 0.83$), ranging from 79.5% on YouTube watch titles to 88.5% on ChatGPT. Topic classification reaches 84.4% overall accuracy ($\kappa = 0.83$), ranging from 78.0% on YouTube watch history to 89.0% on Google Search (Appendix Tables 9 and 10). Residual classification errors were concentrated along the taxonomy's most subjective boundaries, particularly between Concrete facts and Conceptual explanations. Because including the platform name in the prompt could theoretically introduce classifier bias toward platform stereotypes, we conducted an ablation removing the platform indicator across 2,000 requests per platform and show (see Appendix C) that the impact is minimal.

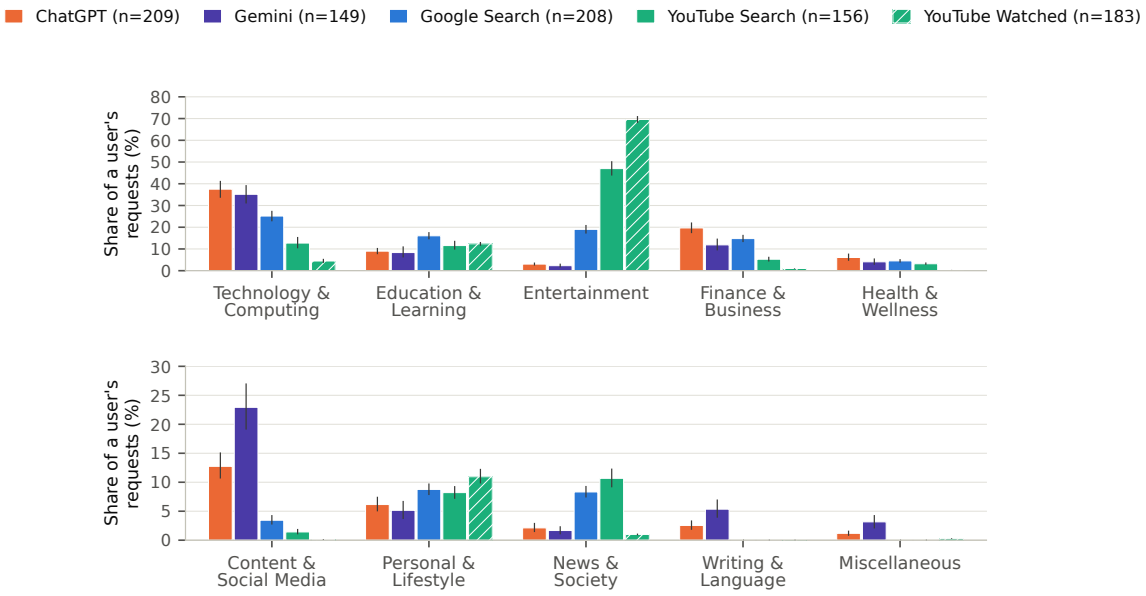


Fig. 1. Topic shares by source (mean per-user share with 95% bootstrap confidence intervals) for the participants. Hatched bars denote YouTube watched video titles rather than user-issued queries. ChatGPT and Gemini are used heavily for computing, content creation, and lifestyle; YouTube is dominated by entertainment; and Google Search is broadly distributed across all topics.

4 Findings

We report four core findings on how information seeking is distributed across platforms. First, we examine how requests divide across surfaces by topic and intent, showing that platforms differ primarily by functional move rather than subject matter (Section 4.1). Second, we show that this functional division persists within specific topical domains (Section 4.2). Third, through qualitative analysis of full user histories, we show how single information-seeking tasks routinely cross platform boundaries (Section 4.3). Finally, we evaluate the longitudinal effect of ChatGPT adoption on Google Search, demonstrating that while overall search volume declines modestly, search composition remains remarkably stable (Section 4.4). Unless stated otherwise, all analyses use the paired cohort of 209 participants over the aligned window (January 2023 to February 2026), reporting bootstrap 95% confidence intervals clustered by user.

4.1 Topic and Intent Distribution Across Platforms

The platforms overlap extensively in subject matter (Figure 1). Technology & Computing is the single largest topic across ChatGPT (37.5%), Gemini (35.2%), and Google Search (25.1%). Education & Learning accounts for 8% to 16% of activity across all services, while Finance & Business represents 19.7% of ChatGPT conversations and 14.8% of Google queries. While specific concentrations exist, such as entertainment dominating YouTube (47.0% of searches, 69.6% of watched videos) and content drafting appearing frequently on conversational AI (12.7% on ChatGPT, 22.9% on Gemini), no topic is exclusive to a single interface.

Instead, platforms diverge sharply in the functional move requested (Figure 2). Three of the eight intents are largely dictated by interface affordances. Unsurprisingly, users do not ask a chatbot to navigate to a website, nor do they expect a search engine to author a finished document. Yet quantifying these boundaries reveals that roughly half of all search

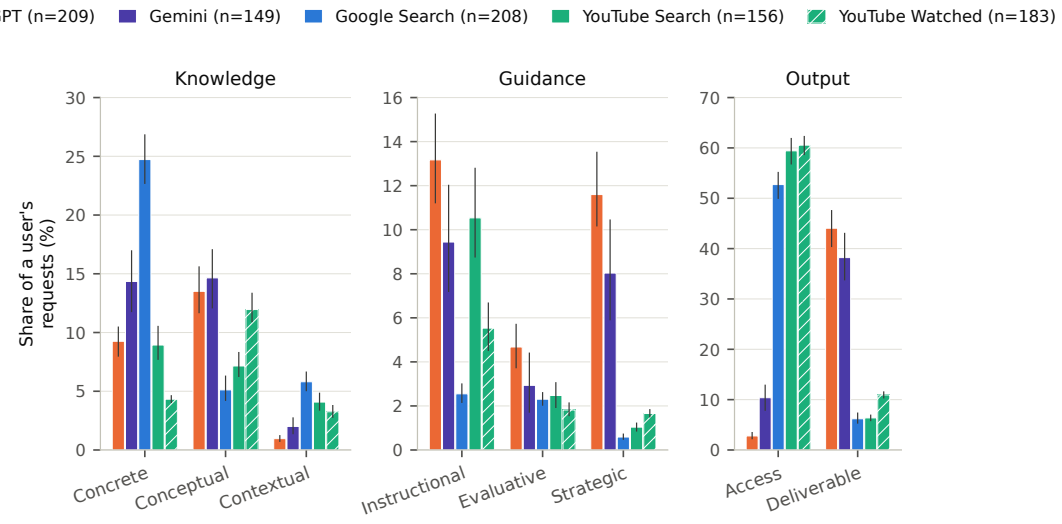


Fig. 2. Intent shares by source across eight functional categories grouped into three families (mean per-user share with 95% bootstrap confidence intervals). Output: Access dominates Google Search and YouTube Search, while Output: Deliverables dominate conversational assistants. Among shared information needs, factual lookups (Concrete knowledge) remain on Google Search, while personalized advice (Strategic guidance) and explanations (Conceptual knowledge) concentrate in ChatGPT and Gemini, and procedural how-to (Instructional guidance) splits between chatbots and YouTube.

activity (52.7% of Google queries and 59.4% of YouTube searches) is devoted strictly to resource access, while over 40% of conversational assistant usage (44.1% on ChatGPT and 38.2% on Gemini) is devoted to generating finished deliverables. Similarly, Contextual knowledge (retrieving real-time or hyper-local data) remains almost entirely with search (5.8% on Google vs. 1.0% on ChatGPT).

These three categories establish the baseline constraints of each system. The more consequential behavioral pattern lies in the remaining five intents, tasks that could plausibly be served by either a search engine or a conversational assistant. Here, users make deliberate routing choices.

The critical empirical pattern emerges among the remaining five intents, which could plausibly be served by any of these systems. Here, users make clear choices between platforms. Concrete factual lookups are mostly done on web search. Verifiable facts and definitions make up 24.7% of Google queries and 8.9% of YouTube searches, compared to 9.3% of ChatGPT requests. In contrast, explanations and advice are mostly done on conversational assistants. Conceptual knowledge accounts for 13.5% of ChatGPT requests versus 5.1% of Google queries, while Strategic guidance (personalized advice based on individual circumstances) comprises 11.6% of ChatGPT and 8.0% of Gemini requests, but only 0.6% of Google queries. Procedural instructions are split between text and video. Instructional guidance accounts for 13.2% of ChatGPT requests and 10.5% of YouTube searches, compared to 2.5% of Google queries. Evaluative guidance represents a small share across all tools (2% to 5%), leaning slightly toward assistants. Across the entire Guidance family, advice and instruction constitute 29.5% of ChatGPT requests but only 5.4% of Google searches.

Routing Choices Within Individual Users. Aggregate platform shares could theoretically arise if different demographics use different tools (for example, if heavy chatbot users simply have different information needs than heavy searchers). To determine whether the same individual routes different tasks to different systems, we analyze routing within users.

For each user u and intent i, we define the routing share $r(u, i)$ as the proportion of that user's intent-i requests that went to ChatGPT rather than Google Search:

$$r(u,i) = \frac{n_{\text{ChatGPT}}(u,i)}{n_{\text{ChatGPT}}(u,i) + n_{\text{Google}}(u,i)} \tag{1}$$

calculated over pairs with at least 20 requests. Because our participants issue roughly twenty Google queries for every ChatGPT conversation, raw routing shares reflect the overall volume disparity between platforms. To isolate relative preference, we also compute the *within-person lift*, defined as the proportion of intent i within a user's ChatGPT requests divided by its proportion within their Google searches:

$$\text{Lift}(u,i) = \frac{p(i \mid u, \text{ChatGPT})}{p(i \mid u, \text{Google})} \tag{2}$$

where a lift of 1 indicates that the user allocates the intent to both platforms in equal proportion.

Figure 3 displays $r(u, i)$ and $\text{Lift}(u, i)$ across 1,074 user-intent pairs. The routing share is highest for Strategic guidance (median 49%), followed by Deliverable (25%), Instructional guidance (21%), and Conceptual knowledge (12%), while dropping near zero for Evaluative guidance (4%), Concrete knowledge (1%), Access, and Contextual knowledge (0%). The within-person lift demonstrates this divergence clearly: for the median user, Strategic guidance carries 14 times the relative weight on ChatGPT that it does on Google, Deliverables carry 5.6 times, Instructional guidance 4.8 times, and Conceptual knowledge 2.7 times. Conversely, Concrete facts carry a lift of only 0.4, Contextual knowledge 0.3, and Access 0.06.

To test these routing preferences formally while controlling for overall individual usage levels, we estimate a conditional logit model predicting platform choice (ChatGPT versus Google) as a function of intent, stratifying by user (Appendix Table 5). Relative to a Concrete fact lookup, the odds that a user directs a request to ChatGPT rather than Google are 36 times higher for Strategic guidance, 15 times higher for Instructional guidance, 14 times higher for a Deliverable, and 8 times higher for Conceptual knowledge, while being 12 times lower for Access. A pooled logit model without individual fixed effects yields nearly identical estimates, confirming that routing is driven by functional intent rather than user-level heterogeneity.

Furthermore, this routing hierarchy is consistent across users. When we rank the eight intents by $r(u, i)$ for each individual, Kendall's coefficient of concordance across the 121 users with activity in all eight categories is $W = 0.67$ ($p < 0.001$). For 81% of these users, the single most assistant-directed intent is either Strategic guidance or Deliverable. While participants vary in their degree of adoption (the interquartile range for Strategic guidance routing spans from 15% to 77%), the directional ordering is shared: when individuals seek personal advice or finished artifacts, they turn to conversational assistants. When they seek verifiable facts or web resources, they remain with search.

Language and Phrasing of Requests. These quantitative categories correspond to distinct expressive styles in the raw text. On ChatGPT, requests focus on creation and problem-solving. Users request programming code ("`write a program to check vowels, consonants, and numbers in Java`"), formal correspondence ("`can you draft a paragraph for an university about education and employment background`"), image modifications ("`please make an image brother`" (translated from Hindi)), and document formatting ("`Can you make this resume professional for me`"). Users frequently request iterative evaluation, such as practicing a mock HR interview prior to an employment screening. The second signature is personalized situational advice, e.g. "`what if i do cfa level 1 and mba. how much salary i can get and does collage matter in it`"; "`how to earn money online using a mobile phone in India`" (translated)); and queries evaluating personal symptoms ("`can i eat pizza my cold cough is`

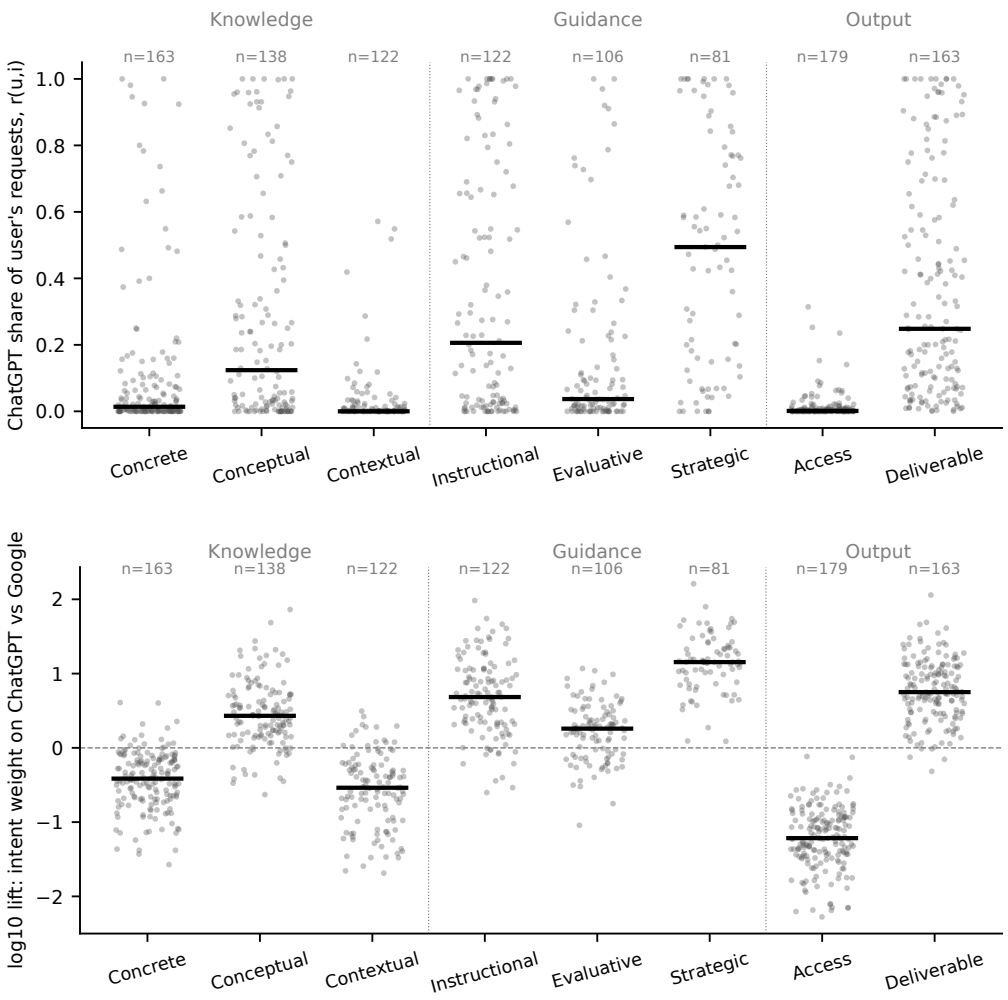


Fig. 3. Within-person routing between ChatGPT and Google Search for the 209 paired participants over user-intent pairs with at least 20 requests. Top: routing share $r(u, i)$, the proportion of a user's intent, i, requests directed to ChatGPT rather than Google. Bottom: within-person lift on a log scale (0 denotes equal representation on both platforms). Solid horizontal bars are the medians.

`70% gone`"). Prompts are conversational, frequently composed in mixed Hindi-English (Hinglish), and address the model affectionately as "`bhai`" or "`yaar`" (brother or friend).[3]

On Google Search, requests are brief, keyword-driven lookups aimed at navigation and retrieval. Access queries name specific destinations: streaming services ("`disney plus hotstar`"), employers ("`infosys`"), e-commerce products ("`kurta pant set`"), or institutional portals. Concrete knowledge queries target rapid verification like exam eligibility criteria, translation of terms into Hindi, sports statistics, or daily market indicators ("`gold rate today`"). Contextual searches are grounded in immediate time and geography, such as local weather, regional news, or university admission

[3]The exact queries through out this paper have been rephrased to avoid reidentification.

Topic	ChatGPT	Gemini	Google Search	YouTube Search	YouTube Watched
Technology & Computing	Deliverable (47%)	Access (21%)	Access (65%)	Access (34%)	Access (44%)
Education & Learning	Deliverable (38%)	Conceptual (31%)	Access (41%)	Access (39%)	Access (72%)
Entertainment	Deliverable (46%)	Concrete (42%)	Access (55%)	Access (82%)	Access (59%)
Finance & Business	Deliverable (37%)	Deliverable (30%)	Access (58%)	Access (45%)	Access (53%)
Health & Wellness	Conceptual (26%)	Conceptual (37%)	Access (35%)	Access (34%)	Access (58%)
Content & Social Media	Deliverable (62%)	Deliverable (86%)	Access (84%)	Access (83%)	Access (70%)
Personal & Lifestyle	Deliverable (31%)	Deliverable (34%)	Access (49%)	Access (45%)	Access (63%)
News & Society	Deliverable (29%)	Concrete (27%)	Concrete (41%)	Access (48%)	Access (80%)
Writing & Language	Deliverable (41%)	Concrete (55%)	n/a	n/a	n/a
Miscellaneous	Deliverable (33%)	Concrete (27%)	n/a	n/a	Access (52%)

Table 2. Plurality intent within each topic and source, with its mean per-user share within that topic, for the 209 paired participants from January 2023 to February 2026. Slices with fewer than 20 requests are omitted; n/a denotes categories structurally absent from search classification (Section 3.2).

dates. Even when users bring advice-seeking needs to Google, they compress them into search syntax, e.g. "`career options after 12th science pcb with high salary in goa`" or "`i got stapler pin in my thumb will it be okay`". Deliverables appear on Google (6.2%) almost exclusively as download queries, where users seek to locate an existing artifact rather than having one authored.

On YouTube Search, requests center on visual demonstrations and specific media channels. Access dominates (59.4%) as users search for named music, daily serial episodes, or educational creators (e.g., "`studyiq`", "`pawan wagh academy`"). Instructional queries represent the second major category (10.5%), targeting physical tasks and tutorials: "`how to fix back posture`", "`how to make resume for job`", and "`puppeteer tutorial`". Across platforms, the same underlying need takes different forms: a health question appears as a personal situation on ChatGPT, a medication name on Google, and a visual tutorial on YouTube.

4.2 Roles Persist Within Topics

To verify that cross-platform differences are not an artifact of topic distributions, we examine intent distributions within each topic (Table 2). Across almost all topics, the functional specialization remains identical. Deliverables constitute the plurality intent on ChatGPT for nine out of ten topics, while Access is the plurality intent on Google Search for seven out of eight populated topics (with Concrete knowledge leading for News & Society). Access also dominates YouTube Search across all eight topics, though in Technology & Computing, Instructional how-to queries run a close second (32% vs. 34%).

Examining specific topics illustrates this persistent division across platforms. In Technology & Computing, users bring coding problems or error logs to ChatGPT expecting working code (Deliverable 46.7%), while turning to Google Search to locate documentation, downloads, or logins (Access 65.2%), and to YouTube for setup tutorials. In Health & Wellness, users consult ChatGPT with personal medical contexts seeking explanations or advice (Conceptual 26.5%, Instructional 19.6%, Strategic 13.2%), such as evaluating home remedies or persistent symptoms. On Google, they search for clinical drug names or specific conditions (such as "`dyslipidemia`" or "`betadine gargle`"), and on YouTube they seek procedural demonstrations (such as "`how to make neem leaves paste for hair`"). In Finance & Business, participants use ChatGPT to formulate trading strategies or career plans, Google Search to check live rates (such as "`gold rate today`") or banking portals, and YouTube for financial explainers. Regardless of the topic, users consistently

treat ChatGPT as an engine for synthesis and advice, Google as an index to destinations and facts, and YouTube as a library of visual procedures.

4.3 How Single Information Tasks Span Multiple Platforms

Quantitative distributions show where categories of requests land, but they cannot reveal how a user coordinates multiple systems during a single task. To observe cross-platform transitions directly, we conducted a qualitative close reading of full user histories.

From the 75 adopters with longitudinal Takeout archives, we purposively sampled fifteen participants across the spectrum of post-adoption search changes. We sampled the five participants with the largest increase in Google Search volume, the five with the largest decrease, and the five closest to zero change. For each participant, the first author examined their complete ChatGPT conversations and Google queries chronologically from the date of their first conversation onward. We identified and coded every instance where searches and conversations within a three-day window belonged to the same overarching information task.

Complementary Workflows. Across these trajectories, participants routinely used ChatGPT to formulate plans and generate drafts, while using Google Search to locate specific tools, primary sources, or external documentation to execute those plans. For example, one participant spent an evening prompting ChatGPT across multiple turns to design a customized seven-day Jain meal plan with explicit macronutrient and micronutrient targets. Over the same week, their Google queries included "`The Encyclopedia of Modern Bodybuilding PDF`", "`40 40 20 vegetarian meal plan`", and "`food database with calories`". The underlying goal about structuring a nutrition and fitness regimen was identical. ChatGPT provided an interactive artifact tailored to dietary constraints, while Google Search supplied the external reference databases and manuals needed to verify it.

We observed this cross-tool interleaving across a wide range of daily activities. In career planning, one participant building a freelance graphic-design career asked ChatGPT which technical skills to highlight, how to master logo design, and how to draft professional objectives. During the same days, they searched Google for Behance, Pexels, Unsplash, Adobe Illustrator downloads, Internshala, and Upwork. After generating a career summary in ChatGPT, they pasted a modified version into Google to compare it against existing online profiles, before returning to ChatGPT to generate client proposals.

Technical development followed a similar pattern. Another participant engaged in an extended multi-day dialogue with ChatGPT to write and debug custom WooCommerce and WordPress plugins. Simultaneously, their Google queries targeted Google Cloud Console, web hosting portals, and the WordPress plugin directory. The assistant provided the programming logic, while the search engine provided access to the execution environments.

For content creation, a participant developing an educational YouTube channel used ChatGPT to draft video scripts and outline scientific topics, while using Google to find automated text-to-speech tools, video editors, and royalty-free graphics.

The same dynamic appeared in personal reflection. A participant held reflective conversations with ChatGPT regarding emotional guilt and interpersonal conflict, while their subsequent Google searches sought out teachings and lectures from specific spiritual leaders. In this case, ChatGPT served as an interactive sounding board, while search connected the user to authoritative external voices.

We also examined histories for evidence of direct substitution. In cases where overall search volume declined, we observed localized instances where quick factual queries and definitions that previously appeared in Google search logs

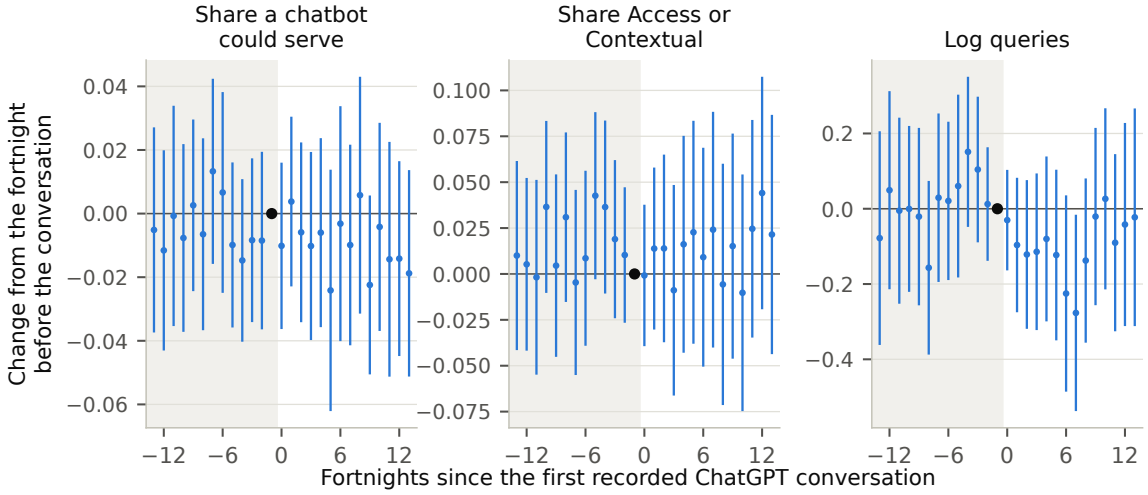


Fig. 4. Event-study estimates of Google Search around an individual's first recorded ChatGPT conversation, for 102 adopters against 74 comparison participants, with user and calendar-fortnight fixed effects and 95% confidence intervals clustered by user. Each point is one fortnight, a two-week period, so the axis covers the 26 weeks before and the 26 weeks after that conversation. Zero is the fortnight containing the first conversation in the donated export, which for some adopters postdates their first use of ChatGPT. The shaded area is the pre-adoption period and the black point at $t = -1$ is the reference fortnight all others are measured against. The panels show the share of queries a chatbot could serve, the share of Access and Contextual queries, and the log query count; Table 8 gives the before-and-after estimates.

disappeared, replaced by direct prompts in ChatGPT. However, we also observed that temporal proximity alone does not imply task integration: users frequently interleave unrelated queries across tabs within minutes of each other.

Crucially, because current interfaces lack native cross-tool integration, users manage this coordination manually, copying tool names recommended by ChatGPT into Google Search, or pasting text discovered via search into ChatGPT for explanation. These qualitative traces illustrate why query counts alone cannot measure technology adoption. Adopting an assistant displaces certain quick searches while generating new, downstream search tasks required to enact the assistant's advice.

4.4 Search Composition and Volume Around Adoption

A prominent narrative in both the technology industry and recent empirical research contends that conversational AI will substitute for traditional web search [1, 9, 31]. Under this substitution hypothesis, adopting an assistant should depress overall search traffic and should specifically siphon away the queries that chatbots handle best, namely conceptual explanations and personal advice. However, our qualitative findings in the preceding section suggest a more complementary arrangement: assistants often generate new, downstream search needs rather than eliminating them. To test whether adoption leads to displacement in practice, we evaluate how individuals' Google Search volume and query composition changed after they began using ChatGPT.

Assessing the impact of adoption in observational logs is complicated by the fact that real-world use offers no unexposed control group. Among the 80 participants who donated only Google archives, 70 showed active searches for ChatGPT, visits to "`chatgpt.com`", or Gemini usage in their Takeout logs (Appendix Table 7). Because non-donors were active AI users, constructing a synthetic control from them would compare AI adopters against other AI adopters. We

therefore conduct an event study tracking 102 adopters relative to their own pre-adoption baseline across a 52-week window (26 weeks before to 26 weeks after each user's first conversation), using user and calendar fixed effects to absorb common secular shocks (formal econometric specification and regression estimates in Appendix D).

The results demonstrate that adopting an assistant reorganizes search workflows without dismantling the underlying composition of web search (Figure 4). Prior to adoption, 12.0% of an adopter's Google queries were for categories a chatbot could serve (Conceptual explanations and Guidance). Following adoption, this share moved by only −0.4 percentage points (95% CI [−1.8%, +0.9%], $p = 0.53$), with no detectable pre-trend ($p = 0.95$). Similarly, the share of Access and Contextual queries remained virtually flat (changing by −0.4 points, $p = 0.79$). At the same time, overall search volume experienced a modest downward drift: query counts fell by approximately 11% ($\beta = -0.117$, $p = 0.07$).

Read together, these findings provide little evidence for the selective substitution hypothesis. While adopters issue slightly fewer searches overall, they bring the exact same functional mixture of requests to Google as they did before adoption. Chatbots did not selectively siphon away explanatory or advice-seeking queries. Google maintains its role as an everyday index for navigation, verification, and web access.

5 Discussion

5.1 The Fragmentation of the Search Task

HCI literature has long established that system affordances shape how users formulate their needs. People compress a health worry into a drug name keyword for Google, while expanding the same worry into a paragraph when requesting judgment from ChatGPT. Even the concept of a deliverable flips depending on the platform. On a search engine, it means the user wants to download a file, but on an assistant, it means they want the system to generate an original artifact. People write their requests specifically for the answer they expect a system to give.

While this behavioral adaptation is expected, it carries two consequences for how we study the impact of AI on search. First, it reveals a measurement trap. Any intent label attached to a log only measures the need as the user formulated it for that specific system. It does not measure the underlying human need. When recent studies conclude that most ChatGPT requests could not be answered by a search engine [44], they are partly measuring this formulation effect. If a user had routed that same underlying need to search, they would have translated it into a query that search could answer.

Second, it shows that foundational models of information seeking have split across platforms. Marchionini separated simple lookup from the exploratory search that leads to understanding [30]. Historically, both happened within the search engine. In our data, that distinction now forms a boundary between platforms. Lookup stays with the search index, while the drive for understanding moves to the conversational assistant. Bates described a search process in which a user reformulates their query as each new result changes what they need [2]. Today, that iterative path crosses system boundaries, leaving each platform's log with a fragment of the task.

5.2 Conversational AI Serves Unmet Needs

Media niche theory predicts that a new medium displaces an old one only when both serve identical gratifications. When they offer different utilities, they differentiate and coexist [12, 13]. The platform choices we observe fit this prediction, and they also show where the new work came from. The two largest categories on ChatGPT are finished artifacts and personal advice. Neither category had a home in search. Deliverables reach Google almost exclusively as download queries, and requests for situated advice account for less than one percent of Google traffic. Before conversational

assistants, an artifact was made by the person or paid for, and advice came from a senior colleague, a tutor, a doctor, or a forum. For a young population working toward examinations and income, often without ready access to mentors, the assistant's nearest competitor for these tasks is a human or the user's own labor, rather than a search engine. Prior donation studies in the Global South found personal problem-solving outweighing professional work [10]. Our cross-platform view adds that this work was not taken from search.

Among the five contested intents, the pattern is differentiation. Users route factual queries to the search index and direct requests for explanation or personal advice to the assistant, while turning to both the assistant and video platforms for step-by-step instruction. The desired form of the answer separates them. A fact requires a source that can be verified. Advice requires a synthesis fitted to the asker. The exposure of search to the assistant is therefore bounded. Before adoption, about one in eight of an adopter's Google queries asked for something a chatbot could serve. Even complete displacement of those queries would leave most of what search does for these users untouched. The substitution debate imagines the assistant contesting the center of search. For this cohort, the center of search is reaching named places and checking facts, and the assistant contests its edge.

Users agree on which system suits a given task and differ widely in how much of it they send there. For advice-seeking, a quarter of participants send less than a sixth of it to ChatGPT, while another quarter send more than three quarters. The repertoire literature characterizes these as different compositions of a single set of channels [17, 29]. Describing these types, who has moved their reasoning to the assistant and who uses it only for production, is the next descriptive task.

5.3 Manual Handoffs and Cross-Platform Verification

The histories show a recurring sequence of manual handoffs. A need is articulated to the assistant as a situation and comes back as a plan, a draft, or a list of named things. The user then reaches those named things on the web, finding the tools, sources, platforms, and downloads that the plan mentioned. Sometimes a page found this way is carried back into the conversation to be explained. Sometimes the assistant's output is typed into Google to see how it compares with what others wrote. The assistant creates search work, because a plan is a list of things to find on the web. This is one reason adoption can leave the composition of search unchanged. Some moves are displaced, others are created, and the mix stays where it was. Channel complementarity theory predicts that bringing a domain to a new channel goes with using the old channel more for it [14]. Our histories read that way. The event study did not test it by domain, so it stays a hypothesis for the next study.

The direction of these handoffs depends on what the user is trying to accomplish. Panel work in the United States and the United Kingdom found that cross-surface sessions more often begin in search and move to the assistant [18]. Our histories show movement in both directions. Search comes first when a lookup turns into a question. The assistant comes first when the person has something to build, and they enter the web through the plan.

These friction points are places where design could help. Rather than merging the two systems into one, each could support the handoffs. An assistant's plan could be exported as a set of editable web searches, so that the plan's structure survives the move to the web. Because users rely on search to verify the assistant, any system that merges the two functions, such as an assistant that browses the web, should keep the verification visible, showing what was compared against what and where the sources disagreed.

5.4 Seeking Advice from Systems Without Sources

The stakes of these platform choices are high. A search engine returns sources and leaves synthesis and credibility judgments entirely to the user. An assistant returns a single authoritative voice and shows little of where it came from. People send personal advice and explanation requests to the single voice, discussing health, careers, money, and relationships. These are the domains where no ground truth exists to check an answer against, and where accountability matters most. They keep factual queries, where a wrong answer is cheap to detect, with the system that shows its sources. As a matter of functional fit, this makes sense. An index cannot tell a user whether to take a particular career path or eat pizza with a bad cough. The consequence is that provenance is available where it is least needed and absent where it is most needed.

Two findings sharpen the point. A quarter of participants send more than three quarters of their advice-seeking to ChatGPT. For them, the assistant is the main channel for judgment about their own lives. The register of the requests points the same way. The use of Hindi and romanized Hindi, kin terms such as bhai and yaar, and symptoms described in the first person show that the assistant is addressed as a confidant rather than consulted as a reference. Donation studies of ChatGPT report this same framing and a similar growth in personal requests [10, 22]. Critics of conversational search warned that a single synthesized answer removes the cues people use to judge credibility [35], and opinionated systems narrow what people encounter [37]. Our data show which requests reach that format. Users route the requests where credibility cues matter most to the system that hides them.

The obligations of an assistant should follow how it is used. In this population it is a guidance channel, whatever it was built as. As a design implication, conversational assistants should recognize when they are fielding high-stakes advice or time-sensitive factual queries. Instead of answering these requests in a synthesized voice stripped of provenance, the system should hand the move back to a source, routing users toward professional resources or a person when appropriate.

5.5 Measuring What Generative AI Does to Search

The debate over whether AI will replace search relies on the metrics these systems produce. Search engines count queries, chatbots count prompts, and panels count visits. The question became whether one count falls as the other rises. Total query volume can hold while the content of the queries changes. Volume can also drop for reasons unrelated to what a chatbot can do. In our data, volume drifted down by roughly a tenth while the composition did not move. An even decline across kinds of request is not what substitution looks like.

Studies with access to content should report the composition of search beside its volume and condition the outcome on intent, or on a proxy for whether a chatbot could have served the request. Studies without content should say which shifts they cannot see. The conflicting panel results currently in the literature, ranging from a decline [31] to a temporary rise [15] to a shift toward leisure [3], can all be true of populations with different mixes of moves. Each measures a sum whose parts move differently. A null finding also needs its interval stated. Our results exclude a fall of more than fifteen percent in the chatbot-servable share over six months, but we cannot exclude a fall of a fifth in total volume.

Data donation made visible two problems that apply to every study of AI adoption. First, there was no AI-free comparison group in this cohort. Seven in eight participants who donated no ChatGPT archive had used ChatGPT or Gemini, and their own Google logs showed it. A control group defined by not installing an app or not subscribing is a claim about behavior that only behavioral logs can verify. A study without content would have called these participants

a control group. Second, the adoption date has the same problem. A third of adopters with a recorded visit to ChatGPT had made that visit more than a month before their first surviving conversation. Exports are censored by retention settings, deletions, and logged-out use. Data donation is costly, self-selected, and censored in ways the platforms control. It is also a design in which these problems can be seen and partly diagnosed from the same data.

6 Limitations

Our findings are not representative of a wider public. Our sample consists of predominantly male Indian platform workers who self-selected to donate data. Furthermore, platform retention settings censored the pre-adoption search history for most users, restricting our quantitative adoption analysis to a subset of 102 people. The language model labels agree with one author's coding on about 85 percent of items, and intent labels measure needs as they are formulated for a specific system, which cannot recover the underlying need. Finally, because devices are often shared within households, we excluded watch history from our within-person claims, though we cannot rule out account sharing on other platforms.

Our adoption analysis is observational and lacks an AI-free control group. We can exclude a decline in total search volume larger than a fifth, and a drop in the chatbot-servable share larger than fifteen percent, but our design cannot detect smaller shifts. The manual handoffs we observed between systems were established by close reading. A validated episode-level linkage with a shuffled-time baseline is the natural next study. Finally, the study window sets a scope condition. Google's generative features, such as AI Overviews, were not broadly available in India during this period. The search engine in these logs is the classical index, so the two surfaces were separate. As integrated AI summaries roll out globally, users may shift some conceptual requests to the results page rather than routing them to a standalone assistant [1].

7 Conclusion

We compared what the same Indian users brought to ChatGPT, Gemini, Google Search, and YouTube over three years of donated logs. The platforms differ less in subject than in the tasks users ask of them. People route artifacts and personal advice to the assistant, factual lookup and navigational reach to the search engine, and requests for visual demonstration to video platforms. The pattern holds within individuals, persists across topics, and appears in whole histories as handoffs between systems inside single tasks.

Following a person's first ChatGPT conversation, the composition of their Google searches does not change, and the count drifts down by about a tenth. A query is not the task. The assistant's largest uses are work that never had a home in search, and its contest with search happens at the edges of what search does. The work between these systems is done by hand today, and that friction is where verification happens. That is where design and measurement should look next.

Use of Generative AI

Language models were used in this work in two ways. (i) As research instruments, open-weight models ran on institutional hardware to embed and cluster requests and to assign topic and intent labels (Section 3); their labels were checked against human coding (Appendix E). (ii) As writing and programming aids, the authors used Anthropic's Claude and Google's Gemini to revise prose in every section of this paper, to write and check the analysis and plotting code, and to review drafts. The research questions, study design, data collection, final write up and interpretation are

the authors' own. The authors checked the code and analysis done by the AI models, and take full responsibility for the content.

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A Participant Cohorts

Because different analyses require different combinations of platform archives and history lengths, the exact number of participants varies across the paper. Table 3 summarizes the specific cohort sizes used for each analysis.

Cohort Requirement	Participants	Used In
Donated both ChatGPT and Google Search logs	209	Main comparisons
Donated all five platform logs	148	Robustness checks
Sufficient Google history for adoption analysis	102	Adoption analysis
Did not donate a ChatGPT archive	80	Contamination check

Table 3. Number of participants satisfying the requirements for different analyses.

B Topic Classification

B.1 Topic Modelling Parameters

Table 4 lists the BERTopic parameters selected by Optuna for each conversational source.

Parameter	ChatGPT	Gemini	YouTube Watched
UMAP			
Number of components	3	5	8
Number of neighbours	8	8	37
Minimum distance	0.0167	0.0673	0.4860
HDBSCAN			
Minimum cluster size	47	10	45
Minimum samples	5	3	25

Table 4. BERTopic parameters selected per source.

B.2 Google Trends Classification Prompt

Google Search and YouTube Search queries are classified in two stages with the prompt below: first into one of the 24 top-level Google Trends categories, then into a subcategory conditioned on the first result. The full two-level taxonomy is the public Google Trends category list.[4]

System Prompt: Search Query Topic Classifier

You are an expert classifier for Google search queries. Given a search query, choose exactly ONE topic from the provided options. You must always return a valid label from the list. Answer with the label string only, no explanations or extra text.

User:

Here is a Google search query:

`{query}`

Allowed topics:

`{options}`

Choose the single most appropriate topic. Return only the label, exactly as it appears in the list above.

B.3 Within-Person Routing Models

Table 5 gives the conditional-logit and pooled-logit estimates summarized in Section 4.1.

C Intent Classification

C.1 Classifier Prompt

The prompt below is used for every source. The Source field is present in the version reported in the main text; the masked re-run of Section 3.3 removes it.

[4] https://github.com/pat310/google-trends-api/wiki/Google-Trends-Categories

Intent	Within-user coef.	(SE)	OR	Pooled coef.	(SE)	OR
Concrete (reference)	0		1.00	0		1.00
Conceptual	2.10	(0.03)	8.17	1.93	(0.12)	6.89
Contextual	−0.82	(0.05)	0.44	−0.71	(0.17)	0.49
Instructional	2.74	(0.03)	15.47	2.67	(0.15)	14.41
Evaluative	1.36	(0.04)	3.91	1.44	(0.13)	4.21
Strategic	3.59	(0.04)	36.11	3.40	(0.14)	29.91
Access	−2.47	(0.04)	0.08	−2.17	(0.12)	0.11
Deliverable	2.65	(0.03)	14.13	2.53	(0.17)	12.60

Table 5. Log-odds that a request goes to ChatGPT rather than Google Search, by intent, relative to Concrete knowledge; 209 paired participants, 468,141 requests. Within-user: conditional logit with participants as strata. Pooled: ordinary logit with standard errors clustered by participant. Kendall's W over the 121 participants with all eight intents is 0.67 ($p < 0.001$).

User Intent Classifier

You are an expert at classifying user intent from search queries and conversational AI interactions. You will be given a query (from Google Search, YouTube, ChatGPT, Gemini, or a browser history) and must assign it to exactly one of the following 8 intent categories, organised under 3 broad types.

Taxonomy:

1. Knowledge Intent: The user is seeking information: facts, explanations, or situational updates. The system is treated as a source of objective or encyclopedic knowledge.

- **Knowledge: Concrete**: Look-ups for verifiable facts, definitions, or straightforward answers. The answer is discrete and confirmable. *Keywords:* what is; who; when; does; list; define; example of. *Examples:* "highest cloud distance from earth in feet", "what is microbiology", "does bacon have carbs"
- **Knowledge: Conceptual**: Explanations of concepts, mechanisms, or how/why something works. Requires understanding, not just a lookup value. *Keywords:* how does; why; mechanism; difference; explain; summarize. *Examples:* "how does 401K work", "global warming causes", "Difference between pulse rate and heart rate"
- **Knowledge: Contextual**: Information that changes based on time, location, or recency: live data, local results, or current events. *Keywords:* current; latest; near me; weather; news; today; now. *Examples:* "nba scores", "DMV near me", "news today in united states"

2. Guidance Intent: The user wants actionable direction: instructions, comparisons, or strategic advice to help them act or decide. The system is treated as a helpful advisor or expert.

- **Guidance: Instructional**: Step-by-step, actionable guidance for completing a task or solving a concrete issue. *Keywords:* how to; steps; fix; remove; solve. *Examples:* "How to make chicken curry", "how to use an EpiPen", "how to remove burnt odor in house"
- **Guidance: Evaluative**: Guidance focused on assessing, comparing, or weighing alternatives and trade-offs. *Keywords:* best; compare; versus; -est superlatives; suggestion. *Examples:* "Best workouts for increasing flexibility", "Is pork tenderloin better on a grill or in the oven", "Living in Latvia versus US"
- **Guidance: Strategic**: Advice for approaching situations, planning next steps, or generating ideas. Involves broader judgment or personal context. *Keywords:* advice; improve; tips; what to do; should I; ideas; brainstorm; plan. *Examples:* "how to balance work and family life", "should I go to this retirement party", "itinerary with travel dates to Italy"

User Intent Classifier (continued)

3. Output Intent: The user wants a concrete result: access to a resource or a finished artifact. The system is treated as a tool or service provider.

- **Output: Access**: Locating, reaching, or operating existing resources, platforms, or tools for use or interaction. *Keywords:* login; [platform name alone]; book; buy; schedule; calculate; convert. *Examples:* "facebook", "grammarly", "mic test", "Countries flag quiz"
- **Output: Deliverable**: Obtaining a finished artifact: retrieved, transformed, summarised, or newly created content. *Keywords:* download; template; summarize; rephrase; rewrite; write; create; generate. *Examples:* "Update my resume", "write a birthday poem for a 17-year-old", "Can you make this more professional"

Classification Rules:

- Assign exactly one label. If a query fits two categories, choose the one that best reflects the primary intent.
- Distinguish *Knowledge: Concrete* (single verifiable answer) from *Knowledge: Conceptual* (requires explanation).
- Distinguish *Knowledge: Contextual* (answer depends on time/location) from *Knowledge: Concrete* (stable fact).
- Distinguish *Guidance: Instructional* (specific steps) from *Guidance: Strategic* (broader planning or advice).
- Distinguish *Guidance: Evaluative* (comparing options) from *Guidance: Strategic* (deciding what to do in a situation).
- Distinguish *Output: Access* (reaching/using a resource) from *Output: Deliverable* (getting a finished artifact).
- The source field provides platform context but does not change the underlying intent category.

Output Format: Respond with exactly one label. No explanation, no preamble, no punctuation.

User: Classify the following query.

Source: `{source}` **Query:** `{query}`

C.2 Validation Against Human Coding

Tables 9 and 10 (Appendix E) report agreement between the classifier and one author's judgement on label-stratified samples of 200 items per source, for topics and for intents. Table 6 reports how intent labels change when the Source field is removed from the prompt.

Platform	*n* re-labelled	Labels changed
ChatGPT	2,000	7.9%
Google Search	2,000	6.8%
YouTube Search	2,000	16.7%

Table 6. Effect of removing the Source field from the intent prompt.

D Google Search Around Adoption: Setup and Diagnostics

This appendix documents the comparison-group check, the event study of Section 4.4, and the earlier synthetic-control analysis that it replaces.

	Participants	Share
Google archive, no ChatGPT archive	80	
with visits to chatgpt.com or openai.com	45	56.2%
with searches mentioning ChatGPT or OpenAI	44	55.0%
with Gemini activity	64	80.0%
with any of the three	70	87.5%
with none	10	12.5%
Adopters with a Google archive	209	
with a recorded chatgpt.com visit	140	67.0%
first visit more than 30 days before first exported conversation		35.7%

Table 7. Traces of ChatGPT or Gemini use among participants without a ChatGPT archive, and the timing of adopters' first recorded visit to ChatGPT relative to their first exported conversation.

D.1 Comparison Group

Table 7 reports what the Google logs of the 80 participants without a ChatGPT archive contain. Seventy show at least one trace of generative-AI use: 45 visited `chatgpt.com` or `openai.com`, 44 searched for ChatGPT or OpenAI, and 64 have Gemini activity. Ten show none. Among the 209 adopters with a Google archive, 140 have a recorded visit to `chatgpt.com`; the first visit falls a median of 0 days from the first exported conversation, with an interquartile range from 85 days before to 356 days after, and precedes it by more than 30 days for 36 percent of them.

D.2 Event Study

The panel has one row per participant and fortnight from January 2020 onward, restricted to fortnights with at least five Google queries: 3,594 rows from 102 adopters, observed from 26 weeks before to 26 weeks after their first exported conversation, and 74 comparison participants observed throughout. We specify the event study as:

$$Y_{u,t} = \alpha_u + \gamma_t + \sum_{k=-13, k\neq -1}^{13} \beta_k \cdot \mathbf{1}\{t - \text{Adopt}_u = k\} + \varepsilon_{u,t} \tag{3}$$

where α_u are participant fixed effects, γ_t are calendar-fortnight fixed effects, and k denotes the fortnight relative to adoption (with $k = -1$ as the reference period). Standard errors are clustered by participant. We evaluate three dependent variables $Y_{u,t}$: (i) the proportion of queries a chatbot could serve (Conceptual knowledge and Guidance); (ii) the proportion of Access and Contextual queries; and (iii) overall query volume measured as $\log(1 + \text{queries})$.

Table 8 reports the pooled before-and-after differences, and Figure 4 in the main text displays the fortnight-by-fortnight coefficients β_k. Wald tests of the pre-adoption coefficients ($\beta_{k<-1} = 0$) show no pre-existing trends, yielding $p = 0.95$ for the chatbot-servable share, 0.79 for the Access and Contextual share, and 0.75 for the log count. Because the comparison participants also use generative AI, these estimates capture within-person shifts net of calendar trends among AI users, rather than causal effects relative to a non-adopting public.

E Manual Verification of Topic and Intent Labels

An author of this work assessed the LLM-assigned labels on fixed, label-stratified samples of 200 items per source for each classification task.

Outcome	Pre-Adoption Mean	Post-Adoption Change	(SE)	p
Chatbot-servable share	0.120	−0.004	(0.007)	0.53
Access / Contextual share	0.565	−0.004	(0.015)	0.79
log(1 + queries)	4.03	−0.117	(0.064)	0.07

Table 8. Estimated change in Google Search behavior during the 26 weeks following ChatGPT adoption relative to the 26 weeks prior, net of calendar-fortnight fixed effects (102 adopters, 74 comparison participants, 3,594 user-fortnights; standard errors clustered by participant).

Source	N	Accuracy	Cohen's κ	Krippendorff's α
ChatGPT	200	83.0%	0.815	0.813
Gemini	200	88.0%	0.868	0.868
Google Search	200	89.0%	0.876	0.876
YouTube Search	200	84.0%	0.821	0.820
YouTube Watched	200	78.0%	0.759	0.756
Overall	**1000**	**84.4%**	**0.828**	**0.827**

Table 9. Manual verification of topic labels across the five sources. Each source contributes 200 reviewed items; the pooled row combines all 1,000 items.

Source	N	Accuracy	Cohen's κ	Krippendorff's α
ChatGPT	200	88.5%	0.871	0.870
Gemini	200	88.0%	0.865	0.865
Google Search	200	84.5%	0.827	0.826
YouTube Search	200	83.0%	0.810	0.809
YouTube Watched	200	79.5%	0.772	0.770
Overall	**1000**	**84.7%**	**0.829**	**0.828**

Table 10. Manual verification of intent labels across the five sources. Each source contributes 200 reviewed items; the pooled row combines all 1,000 items.